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# Project Azalea – Do LLMs Exhibit Gender Bias During Resume Screening?

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## Abstract

Do large language models (LLMs) such as OpenAI’s GPT-5 and Anthropic’s Claude 4.5-Haiku exhibit gender bias during resume screening? Using a reconstructed dataset derived from Dewey’s People and Resume Repository, we tested the effects of modifying gendered information in resumes for LLM-based applicant rankings under different conditions. For each industry and job role combination, the models were instructed to rank sets of fifty randomly sampled entry-level resumes under three scenarios: ground truth (original data), gender-null (gender and interest fields removed), and gender-switched (gender values reversed). Rankings were repeated fifty times across five batches per condition and aggregated using a Borda count to establish consistent baselines. Statistical differences across ranking distributions were evaluated using the difference between mean rankings among batches. Results show that gender-neutral or switched resumes slightly raised women’s rankings and marginally lowered men’s, with more pronounced effects in the hospital and healthcare industry than in computer software. While the overall differences were small (typically one to two ranks), our findings suggest that gender cues can contextually influence LLM behavior, warranting a large-scale replication of this investigation to assess fairness and accountability in automated hiring.

## 1 Introduction

LLMs are increasingly adopted by recruiters as a scalable and efficient screening tool to reduce large applicant pools to a targeted set of candidates for interview. As LLMs begin to supplant earlier algorithmic approaches, their role raises critical questions on fairness and bias. While traditional evaluation systems have been scrutinized for encoding gender and racial disparities, the extent to which LLMs exhibit similar biases remains unknown. To assess whether these models exhibit gender-related bias in this screening role, we evaluated their behavior under a set of controlled test conditions that systematically varied the presence and representation of gender information within resumes. This experimental design enables us to quantify the extent to which gender cues influence the candidate-quality rankings produced by different LLMs.

### 1.1 Motivation

Gender bias in hiring is particularly well-documented, yet relatively little empirical work examines whether modern LLMs exhibit such bias when tasked specifically with ranking candidate resumes. Because ranking decisions directly influence who advances to interview stages, even subtle systematic biases may produce meaningful real-world disparities. This motivates a rigorous, controlled investigation into how LLMs respond to gender cues that are embedded in resumes.

## 1.2 Background and Related Work

The background for this project stems from findings presented in Professor Hilke Schellmann’s *The Algorithm: How AI Decides Who Gets Hired, Monitored, Promoted, and Fired*, which provides a detailed journalistic investigation into the expanding role of AI-driven systems in employment decisions from resume screening to video interviewing, skill assessments, productivity monitoring, and even employee termination. Drawing on interviews with vendors, employers, engineers, and job seekers, Schellmann documents how these systems often operate with limited transparency, weak scientific validation, and significant potential for bias. Importantly, she demonstrates that even when employers believe they are adopting neutral or efficient technologies, the underlying models may embed structural biases inherited from training data or optimization heuristics. Her findings on the risks of delegating high-stakes hiring decisions to opaque automated systems was the primary motivator for our investigation on capturing the extent of bias in LLMs during resume evaluation.

Recent work highlights that algorithmic hiring systems may amplify structural biases encoded in LLMs. Guo et al. (2024) showed that demographic and societal stereotypes can arise from pretraining corpora, alignment procedures, and downstream task formulations. Their analysis further demonstrates that these biases can persist even in ostensibly neutral evaluation settings, highlighting the need for systematic bias assessment and mitigation. This evidence is particularly important for high-stakes domains such as resume screening, where biased model behavior can directly influence access to employment opportunities and compound existing inequities.

Building on this broader concern, Imana, Korolova, and Heidemann (2021) showed that algorithmic systems involved in job advertising can produce gender-skewed outcomes even when ads and targeting criteria are neutral. Using paired job-ad experiments, they distinguish between skew driven by legitimate qualification differences and skew introduced by the platform’s own delivery algorithm, finding that the latter can disproportionately reduce certain groups’ exposure to employment opportunities. Their results highlight how discrimination can arise from platform-level optimization rather than explicit intent. Our study builds on this insight by examining the next stage of the hiring pipeline - resume screening - and testing whether LLMs similarly introduce bias. By holding applicant credentials constant and varying only gender cues, we follow a similar logic to assess whether LLM-generated candidate rankings systematically differ by gender.

## 1.3 Research question

We conducted a systematic evaluation of whether LLMs exhibit bias in resume screening through seeing whether altering or removing gender information from resumes would meaningfully affect how LLMs perceive candidate quality. Specifically, we sought to answer three key research questions:

1. Do LLMs alter their ranking of a candidate when gender information is removed or modified?
2. Does flipping a candidate’s gender systematically improve or worsen their perceived quality relative to the original resume?
3. Are such effects consistent across different models—namely OpenAI’s GPT-5 and Anthropic’s Claude-4.5-Haiku—and across industries with different gender compositions?

We hypothesized that the removal or inversion of gendered information would lead to measurable shifts in candidate rankings, revealing potential biases in how LLMs process social and contextual information. By comparing outcomes across multiple models and industries, we sought to identify whether any observed effects were independent or structural.

## 2 Data and Resume Reconstruction

In order to attain a large resume dataset, we used Dewey’s People and Resume Repository, a large-scale database containing LinkedIn-derived resume information. To prepare the data for model evaluation, we reconstructed each individual’s resume as a unified, stand-alone record. This required merging multiple demographic, education, and employment tables through a series of SQL joins, resulting in a structured dataset suitable for LLM-based ranking experiments. Figure 1 shows this transformation, along with relevant columns from each of the three major feature groups, **person** (birth year, sex, job level, industry, interests, skills, certifications), **education** (education type, dates, degrees, majors, minors), and **experience** (company name, dates, industry, experience level).

To create a controlled comparison setting, we then derived two entry-level subsets from the compressed Dewey dataset: resumes from the Hospital and Healthcare sector and resumes from the Computer Engineering / Computer Software sector. Entry-level resumes were chosen to minimize confounding variation in career seniority and to ensure consistency across candidates.

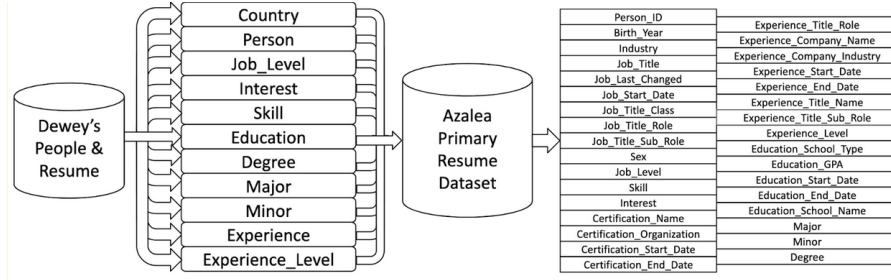


Figure 1: Process of Curating Azalea’s Resume Dataset from Dewey’s People & Resume Repository

### 3 Methodology

#### 3.1 Models Evaluated

We evaluated two LLMs: OpenAI’s GPT-5 and Anthropic’s Claude-Haiku, selected for their practical availability and accessibility and as evidence that our results would reflect models currently integrated into real world hiring tools and workflows, rather than experimental systems.

#### 3.2 Ranking Procedure and Test Conditions

For each industry, both models were evaluated across five randomized batches, each containing fifty resumes. Within each batch, the models ranked resumes under three controlled test conditions. In the **Ground Truth (GT)** condition, the original resume data were provided to the model; in the **Gender Null (GN)** condition, the gender field was removed completely to see how its absence influenced the rankings; in the **Gender Switched (GS)** condition, gender values were reversed (male → female, female → male) to test for directional bias - all of which is illustrated in figure 2. To maintain consistency and reproducibility, all model outputs were requested in a standardized Python dictionary format, allowing for easy parsing and comparison. Each test was repeated fifty times per batch to reduce stochastic variation in LLM responses, resulting in 1,000 rankings per model–industry combination. Figure 3 shows the standardized Python-formatted ranking request used in all experiments; each model received resume data and was instructed to respond only with an ordered list of PERSON\_IDS in ascending order of perceived resume quality.

| Column-Level Experimental Manipulations |                      |                     |                         | Illustrative Row-Level Examples |      |                             |  |
|-----------------------------------------|----------------------|---------------------|-------------------------|---------------------------------|------|-----------------------------|--|
| Column(s)                               | Test 1: Ground Truth | Test 2: Gender NULL | Test 3: Gender Switched | Test 1: Ground Truth            |      |                             |  |
| Sex                                     | Original             | NULL                | Flipped (M ↔ F)         | Person_ID                       | Sex  | Interest                    |  |
| Interest                                | Original             | NULL                | Unchanged               | PERSON_ID_1                     | F    | children, cloud computing   |  |
| Education                               | Original             | Unchanged           | Unchanged               | PERSON_ID_2                     | F    | ice skating, painting       |  |
| Experience                              | Original             | Unchanged           | Unchanged               | PERSON_ID_3                     | M    | hiking, agile methodologies |  |
|                                         |                      |                     |                         | Test 2: Gender NULL             |      |                             |  |
|                                         |                      |                     |                         | Person_ID                       | Sex  | Interest                    |  |
|                                         |                      |                     |                         | PERSON_ID_1                     | NULL | NULL                        |  |
|                                         |                      |                     |                         | PERSON_ID_2                     | NULL | NULL                        |  |
|                                         |                      |                     |                         | PERSON_ID_3                     | NULL | NULL                        |  |
|                                         |                      |                     |                         | Test 3: Gender Switched         |      |                             |  |
|                                         |                      |                     |                         | Person_ID                       | Sex  | Interest                    |  |
|                                         |                      |                     |                         | PERSON_ID_1                     | M    | children, cloud computing   |  |
|                                         |                      |                     |                         | PERSON_ID_2                     | M    | ice skating, painting       |  |
|                                         |                      |                     |                         | PERSON_ID_3                     | F    | hiking, agile methodologies |  |

Figure 2: Column-Level Experimental Manipulations and Illustrative Row-Level Examples

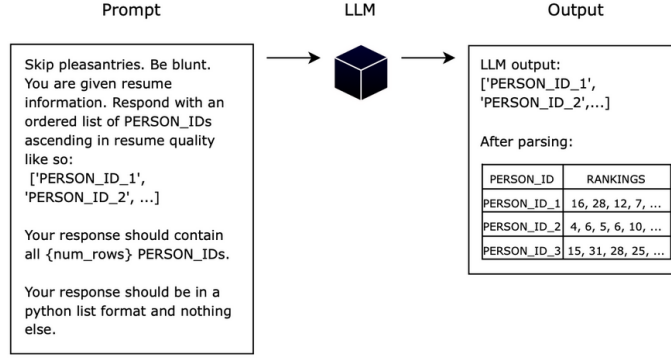


Figure 3: Example of the LLM Evaluation Prompt and Returned Output

### 3.3 Evaluation

For evaluation, we first constructed a filtered resume dataset that selects only candidates within a specific industry, experience level, and role. Once the base dataset is prepared, the pipeline retrieves previously defined (during testing) 50-resume batches from a repository and loads the corresponding ranking results for ground truth (GT), gender-null (GN), and gender-switched (GS) from csv files. These ranking lists are merged back onto the original resume metadata as columns, and all three ranking variants are combined into a single dataframe for each batch. The combined dataset is then split by gender, and for every individual, the code computes GN-GT and GS-GT differences based on average ranking positions. These per-person differences are aggregated across batches and LLM types to form structured heatmap matrices containing results for both male and female candidates under both GN and GS conditions across the five batches for Claude and OpenAI.

## 4 Results

### 4.1 Overall Trends

Both GPT-5 and Claude-Haiku showed relatively stable ranking behavior across test conditions. The average ranking differences typically fell between zero to two positions positive or negative. Despite this, we noticed several subtle gender-linked patterns from the data. On average, women's resumes raised slightly in ranking when gendered information was either removed (Gender Null) or switched (Gender Switched). On the other hand, men's resumes lowered slightly under the same manipulations. These shifts appeared consistently across batches, suggesting that gender cues influenced the model behavior in nuanced ways rather than randomly. For Claude 4.5-Haiku, the effect was more pronounced. There was more variability between conditions, whereas GPT-5 rankings remained less sensitive to gender-related shifts.

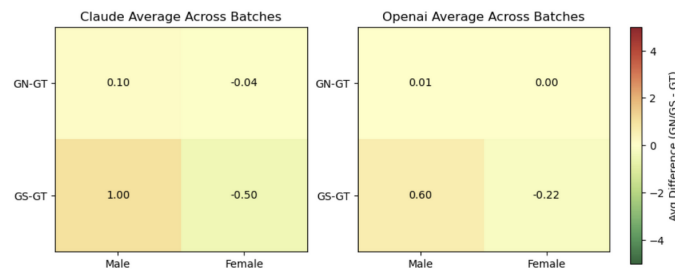


Figure 4: Avg. Overall Diff. Between Gender Null (GN) / Gender Switched (GS) Rankings Compared to Ground Truth (GT) Rankings Across All Batches, for Entry Level Healthcare Workers in the Hospital and Healthcare Industry

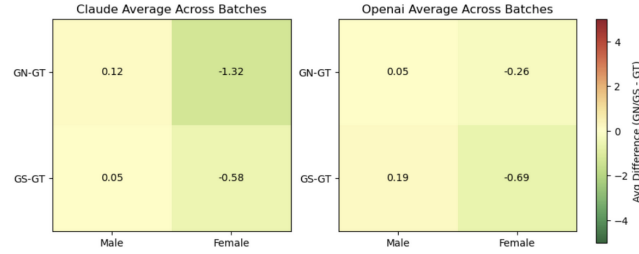


Figure 5: Avg. Overall Diff. Between Gender Null (GN) / Gender Switched (GS) Rankings Compared to Ground Truth (GT) Rankings Across All Batches, for Entry Level Engineers in the Computer Software Industry

## 4.2 Industry Differences

Across both industries tested, men appeared to benefit modestly from being correctly identified as male, while women experienced slight improvements only when gendered information was removed or reversed. In the **Hospital and Healthcare** industry, a traditionally female-dominated field, men’s rankings dropped by roughly 0.6–1 positions when their gender was switched. This pattern suggests that the models may associate certain roles or qualifications with gendered expectations, favoring men when gender information is present. In the **Computer Software** industry, which is a more male-dominated field, the same manipulations produced smaller ranking changes, usually less than 0.25 positions. Thus, while the direction of bias remained consistent—favoring men overall—the magnitude of the effect differed by industry. The evidence indicates that gender cues do, in fact, influence model behavior across domains, but the effects become more pronounced where gender stereotypes are more noticeable.

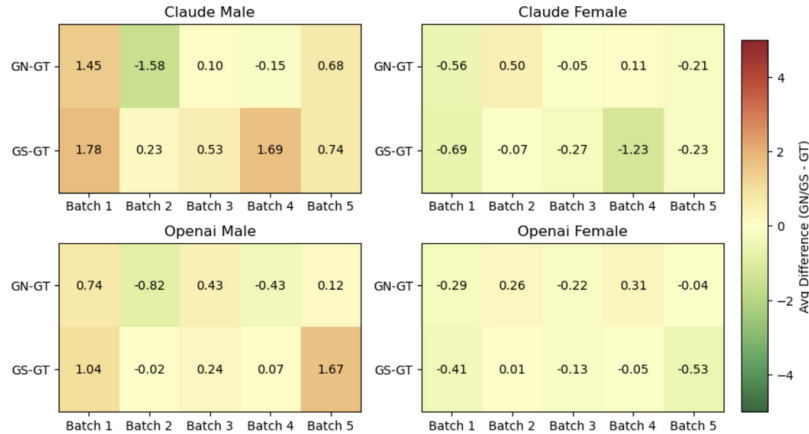


Figure 6: Avg. Diff. Between Gender Null (GN) / Gender Switched (GS) Rankings Compared to Ground Truth (GT) Rankings, Separated by Genders Male (left) and Female (right) and LLM Type Claude (Top) and GPT (Bottom), Among 5 Batches, for Entry Level Healthcare Workers in the Hospital and Healthcare Industry.

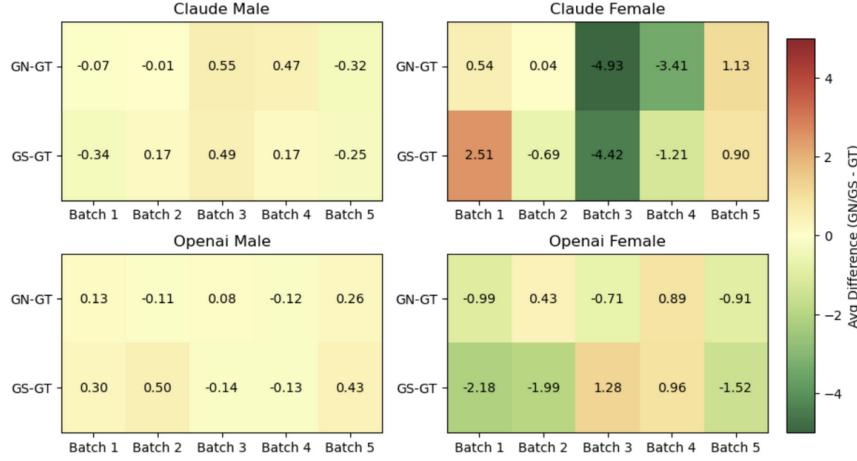


Figure 7: Avg. Diff. Between Gender Null (GN) / Gender Switched (GS) Rankings Compared to Ground Truth (GT) Rankings, Separated by Genders Male (left) and Female (right) and LLM Type Claude (Top) and GPT (Bottom), Among 5 Batches, for Entry Level Engineers in the Computer Software Industry.

### 4.3 Model Comparison

For all our experiments, we found that Claude 4.5-Haiku exhibited higher sensitivity to gender manipulation than GPT-5. Claude 4.5-Haiku also had greater variability, which suggests that its outputs may be more influenced by small changes in contextual cues. In contrast, GPT-5 produced smaller and more stable shifts, showing less reliance on gender features when assessing the quality of a resume. These differences most likely stem from their architectural design: model fine-tuning and training data composition. Although both models showed only minor ranking deviations, their divergent behavior highlights that the outcome can depend significantly on model architecture.

## 5 Discussion/Limitations/Conclusions

Although our results suggest that gender cues can influence resume evaluations produced by LLMs, these findings have to be interpreted with statistical caution. The observed mean ranking shifts fall within the range of expected random variance, and given the limited sample size, industry scope, and the number of repeated simulations, we cannot make any assertions of statistical significance. Instead, our analysis should be understood as a proof of concept for potential bias signals that occur within LLM driven rankings. Future studies should extend this methodology to larger, more diverse datasets and additional social attributes such as race, age, and education. Incorporating explainability metrics (i.e. SHAP values) could further clarify how specific input features drive ranking variation. This study presents a replicable framework for quantifying potential bias in LLM based hiring pipelines.

## Acknowledgments

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